

Development of Force Display Using Pneumatic Actuators for Efficient Conveyance of Emotion

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ABSTRACT

The purpose of this study is to investigate the use of a haptic device to convey emotion through the haptic presentation of gripping. A detailed study into how the intensity of the grip alters the emotions communicated has not been conducted, although gripping is used to convey emotion. In this study, we developed a device for haptic presentation by gripping, and conducted an experiment (N=17) to investigate whether the emotions conveyed by the device changed when the force was varied. The results showed that the magnitude of force, speed, and presentation pattern affected valence and arousal. The results also indicated that not only anger and disgust, but also surprise and happiness can be effectively communicated through this method.

Keywords: Haptic devices, social touch, emotion.

Index Terms: Human-centered computing

1 INTRODUCTION

Implementing a sense of touch in virtual reality (VR) would be useful not only for to reproduce the haptic experience but also to make the experience richer. A previous study showed that adding the haptic information in communication in VR enhanced the abilities of persuasion and leadership [1]. Thus, the use of haptic perception in communication in VR can improve the quality of communication. In this study, we focused on conveying emotion through haptic perception. Previous studies on humans showed that emotion can be conveyed by touch [2], and devices for the purpose have been developed [3, 4]. However, no studies have focused on how the force exerted when gripping conveys emotion and have systematically examined this issue. In this study, we examine what kind of emotion is conveyed when the force is changed in the haptic presentation of gripping. For this purpose, we developed a device that can change the way of grip and evaluate the emotion conveyed by the haptic presentation.

2 DEVICE DESIGN

2.1 Selection of Haptic Presentation Method

In this study, an array of pneumatic actuators was employed to apply force. This is because the method can flexibly vary the magnitude of forces and speeds by controlling airflow amount and airflow rate. In addition to them, we arranged actuators and sequentially inflated them to reproduce the movement of force.

These features allow us to simulate gestures like gripping, stroking and squeezing which are often used to convey emotions[2].

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2.2 Device Overview

Figure 1 shows the developed device. Arduino Nano Every is used to control a valve (SY113-RLZD-PM3-F, SMC) and a motor driver (TB67H450FNG, Toshiba) for managing the flow rate. The actuators, sized at 70 mm × 40 mm, work alongside a flexible force sensor (FSR01CE, Ohmite), adaptable to curved surfaces like an arm, for direct force measurement. Moreover, the flow rate of each actuator can be independently adjusted to control both the magnitude of the force and its speed. This device can thus present varied force exertions.

3 STUDY

An experiment was conducted to evaluate which emotions are conveyed when the force in a grip haptic presentation is changed. Seventeen participants (9 males and 8 females) were involved in the experiment.

3.1 Experimental Procedure

Initially, an experimenter attached a band with three actuators to each participant's non-dominant forearm. The band was then adjusted until the sensor value approached a predetermined level, ensuring consistent band strength for fixation. After that, participants first experienced the maximum force of the presentation before the initial presentation began. To eliminate non-haptic cues, participants wore noise-canceling headphones playing white noise and closed their eyes during the haptic presentation.

Figure 2 shows the actuators' expansion patterns. The experiment used three patterns: All expansion pattern [ALL] (all actuators expand at once, maintaining force for 6 seconds, repeated 3 times), Sequential expansion pattern [SEQ] (actuators expand in sequence, and upon reaching the set force, air flows to the next actuator, 6 expansions per actuator), and All expansion at short intervals pattern [INT] (all actuators expand at once, but unlike ALL the force is maintained for just 0.2 seconds, repeated 6 times). Each expansion pattern was presented in nine different combinations of three force levels (14 N, 20 N, and 29 N for a single actuator) and

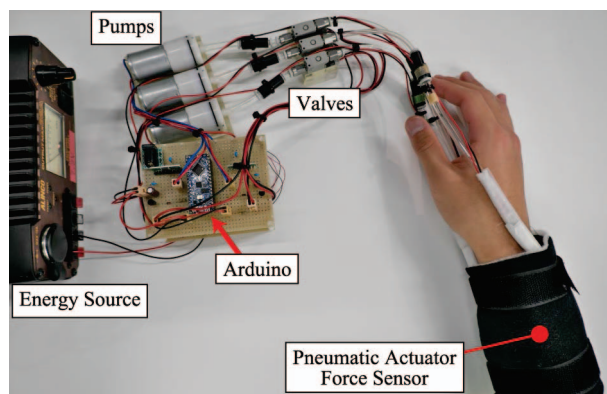


Figure 1: Overview of the haptics device.

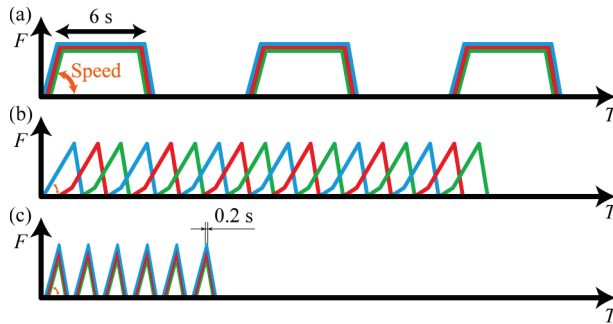


Figure 2: Three types of expansion patterns. Each colored line indicates the magnitude of the force on each actuator. Force magnitude and speed can be changed for all expansion patterns. The magnitude of the force set is the maximum value of the graph, and the velocity is the slope of the graph. (a) All expansion pattern [ALL]. (b) Sequential expansion pattern [SEQ]. (c) All expansion at short intervals pattern [INT]. Note: The scales of the time axis (x-axis) of the graphs do not match.

three speeds (0.71, 1.3, and 2.3 seconds for the time it takes to go from 0 N to 20 N). Each participant experienced 27 presentations, with a 3-minute break after every nine presentations of a single pattern. The order of the nine presentations in each expansion pattern was randomized, and the order of the three expansion patterns was changed so that there were an equal number of trials.

For each presentation, participants were asked "Rate the emotion that you felt the tactile presentation was trying to convey." They rated valence on a 9-point Likert scale (1=Extremely Negative, 5=Neither, 9=Extremely Positive) and arousal on another 9-point scale (1=Extremely Calm, 5=Neither, 9=Extremely Excited). They also rated the six basic emotions (happiness, sadness, fear, disgust, anger, and surprise) on a 7-point Likert scale (1=Strongly Disagree, 5=Neither Agree nor Disagree, 7=Strongly Agree)

3.2 Results

Figure 3 shows the results of the valence and arousal evaluation for the expansion pattern, force, and speed. A within-subjects three-way repeated measures ANOVA was performed on the factors of expansion pattern, force magnitude, and speed. With regard to valence, a significant main effect of forces ($F(2, 32)=74.85$, $p<0.01$) and expansion pattern ($F(2, 32)=9.58$, $p<0.01$) was found. These findings indicate that the greater the force, the more negative emotion is perceived, and SEQ tends to be perceived as a positive emotion. In arousal, there were significant main effects of expansion pattern, force magnitude, and speed (all $p<0.01$). There was also a significant interaction between force magnitude and speed ($F(4, 64)=3.67$, $p<0.01$), and the results of the subtests showed that faster speed was associated with excitement in the weak and medium force cases. In the case of slow and medium speed, the feeling of excitement was felt when the force increased. A significant interaction between the expansion pattern and the magnitude of force ($F(4, 64)=4.05$, $p<0.01$) was also observed. In the case of low force, INT was found to convey more excited emotion than the other expansion patterns. In the evaluation of the six basic emotions, high force and fast speed tended to be perceived as anger. High speed in INT was associated with surprise. Happiness was commonly felt with low force in SEQ. Strong force and fast speed convey disgust emotion. Fear and sadness were typically below neutral in all scenarios.

3.3 Discussion

The psychological experiment results indicated that the grip-like haptic presentation effectively communicates negative emotions. Specifically, increasing force levels were associated with the conveyance of negative emotions. Arousal levels were generally

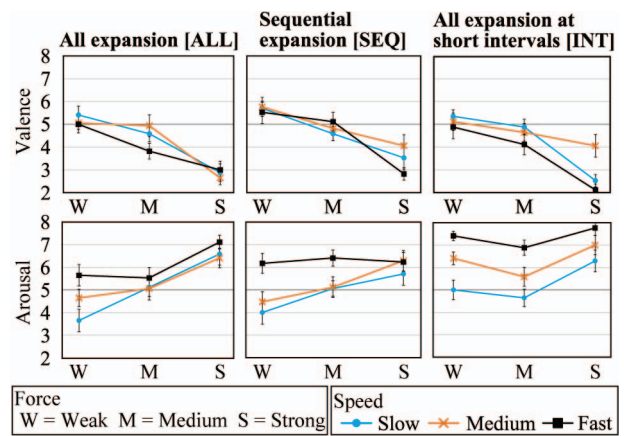


Figure 3: Results of valence and arousal. The graphs indicate group mean plus standard error. The graphs are, from left to right: ALL, SEQ, INT. [Valence] 1=Extremely negative, 9=Extremely positive [Arousal] 1=Extremely calm, 9=Extremely excited (5: neither)

high with increased force and speed, but in some instances, the average arousal remained below the neutral point of 5. Arousal levels remained constant despite changes in speed at high force levels, and similarly, changes in force did not affect arousal at high speeds. Therefore, when either force or speed is high, the associated excitement is conveyed independently of the other factor. In terms of specific emotions, anger and disgust are easily felt, as they are suitable for the transmission of negative emotions. Anger can be conveyed with strong force and high speed, which is consistent with the results of valence-arousal for negative and excited emotions. Similar results were obtained for disgust. On the other hand, happiness was perceived in SEQ with weak force, where valence tended to be high. Surprise may also be conveyed at a speed that is fast enough to be perceived as excitement, and at a weak force that has a valence near neutral.

4 CONCLUSION

This study developed a gripping sensation device for emotional transmission through haptic presentation and conducted a psychological experiment. The study found that a stronger force often communicates negative emotions. Furthermore, it identified that varying the force's strength, speed, and pattern of expansion can express a range of emotions from calmness to excitement. In terms of specific emotions, not only unpleasant feelings such as anger and disgust, but also happiness and surprise, were conveyed by changing the parameters of the presentation. This study suggests that customizing haptic presentation according to the specific emotions intended to be communicated can improve the online communication experience such as in VR.

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